

LAB EXPERIMENT RECORD

Set 6: Product Sales Analysis

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyse monthly product sales using a grouped bar chart, stacked area chart and table, and to build an interactive dashboard combining all three.

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Procedure

1. Store the monthly sales as a matrix with products as rows and months as columns.
2. Use `barplot()` with `beside = TRUE` to draw the grouped bar chart.
3. Compute cumulative sales with `cumsum` logic and draw the stacked area chart with `polygon()`.
4. Create a data frame with totals and display it as a labelled table using `tableGrob()` from `gridExtra`.
5. Build a Shiny dashboard with product and month filters showing the grouped bar chart, stacked area chart and table, and run it with `shinyApp()`.

Key R Functions Used

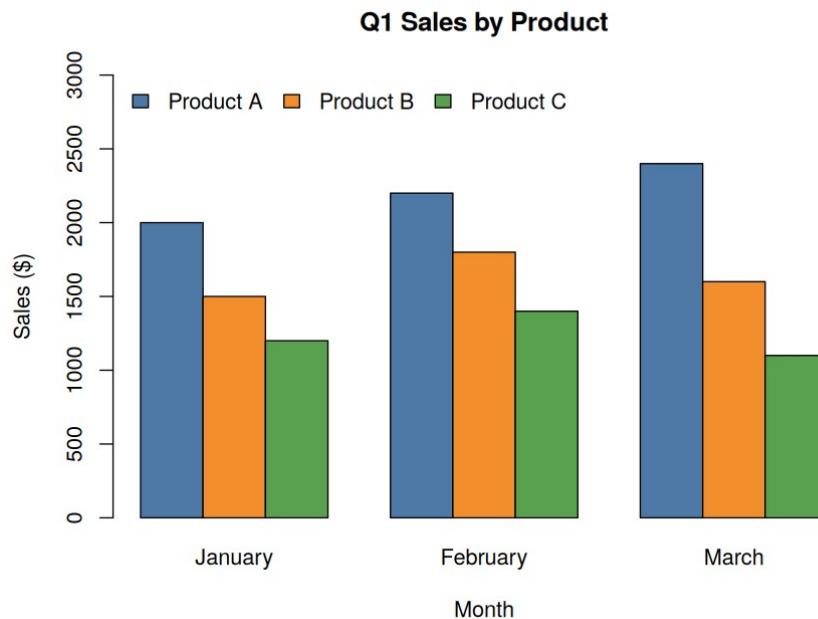
Function	Purpose
<code>matrix()</code>	Stores sales by product and month.
<code>barplot(beside = TRUE)</code>	Creates the grouped bar chart.
<code>polygon()</code> / <code>legend()</code>	Fill the stacked area layers and add a legend.
<code>data.frame()</code> / <code>rowSums()</code>	Build the sales table with totals.
<code>tableGrob()</code> / <code>grid.arrange()</code>	Render the labelled table.
<code>renderPlot()</code> / <code>renderTable()</code> / <code>shinyApp()</code>	Build and run the dashboard.

PROGRAM 1: Grouped Bar Chart of Q1 Sales

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

```
# Dataset
sales <- matrix(c(2000, 2200, 2400,
                  1500, 1800, 1600,
                  1200, 1400, 1100),
               nrow = 3, byrow = TRUE,
               dimnames = list(c("Product A", "Product B", "Product C"),
                              c("January", "February", "March")))
colours <- c("#4E79A7", "#F28E2B", "#59A14F")

# Grouped Bar Chart
barplot(sales,
        beside = TRUE,
        col = colours,
        ylim = c(0, 3000),
        xlab = "Month",
        ylab = "Sales ($)",
        main = "Q1 Sales by Product")
legend("topleft", legend = rownames(sales), fill = colours,
      bty = "n", horiz = TRUE)
```



Output 1: Q1 Sales by Product (Grouped Bar Chart)

Observation: Product A leads every month and grows steadily (2,000 → 2,200 → 2,400). Product B peaks in February (1,800) and Product C also peaks in February (1,400) before falling to its lowest value in March (1,100).

PROGRAM 2: Stacked Area Chart of Overall Sales Trend

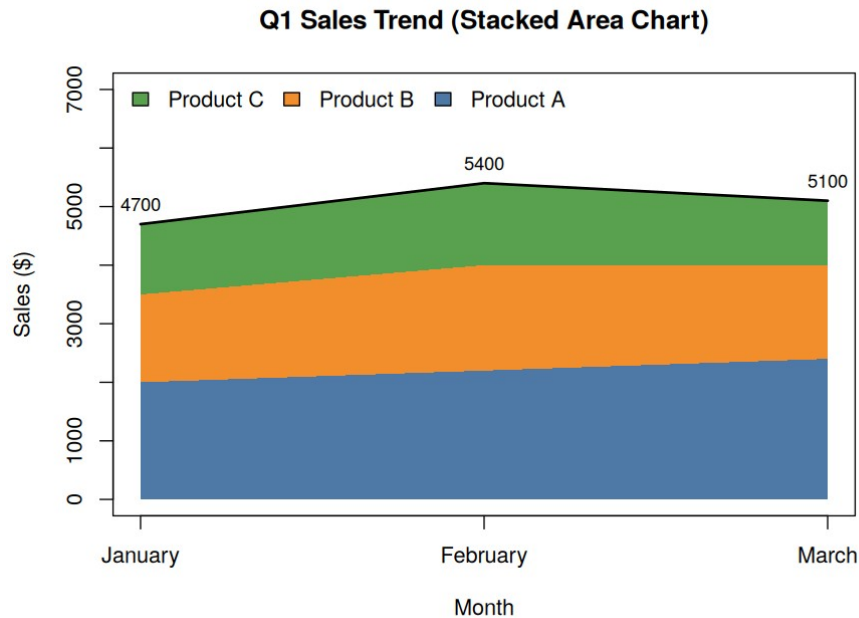
Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
# Cumulative layers
cumA <- sales["Product A", ]
cumB <- cumA + sales["Product B", ]
cumC <- cumB + sales["Product C", ]
m     <- 1:3

# Empty plot area
plot(m, cumC, type = "n", ylim = c(0, 7000), xaxt = "n",
     xlab = "Month", ylab = "Sales ($) ",
     main = "Q1 Sales Trend (Stacked Area Chart)")
axis(1, at = m, labels = colnames(sales))

# Stacked areas
polygon(c(m, rev(m)), c(cumC, rev(cumB)), col = colours[3], border = NA)
polygon(c(m, rev(m)), c(cumB, rev(cumA)), col = colours[2], border = NA)
polygon(c(m, rev(m)), c(cumA, rep(0, 3)), col = colours[1], border = NA)

# Total line with labels
lines(m, cumC, lwd = 2)
text(m, cumC, labels = cumC, pos = 3, cex = 0.85)
legend("topleft", legend = rev(rownames(sales)), fill = rev(colours),
     bty = "n", horiz = TRUE)
```



Output 2: Q1 Sales Trend (Stacked Area Chart)

Observation: Total sales rise from \$4,700 in January to a peak of \$5,400 in February, then dip to \$5,100 in March. Product A forms the largest layer throughout and is the only product still growing in March.

PROGRAM 3: Table of Monthly Sales Data

Build a table to show the monthly sales data for each product. Label the table elements.

```
library(gridExtra)
library(grid)

# Build the table with totals
sales_table <- data.frame(Product_ID = 1:3,
                          Product     = rownames(sales),
                          January     = sales[, 1],
                          February    = sales[, 2],
                          March       = sales[, 3],
                          Total       = rowSums(sales),
                          row.names   = NULL)

sales_table <- rbind(sales_table,
                    data.frame(Product_ID = "", Product = "Total",
                              January = sum(sales[, 1]),
                              February = sum(sales[, 2]),
                              March = sum(sales[, 3]),
                              Total = sum(sales)))

print(sales_table)

# Labelled table graphic
tt <- ttheme_default(base_size = 15,
                    colhead = list(fg_params = list(col = "white", fontface = "bold"),
                                   bg_params = list(fill = "#2F5597")),
                    core    = list(bg_params = list(fill = c("#F2F2F2", "white"))))
title <- textGrob("Monthly Sales Data by Product (Q1)",
                 gp = gpar(fontsize = 13, fontface = "bold"))
grid.arrange(title, tableGrob(sales_table, rows = NULL, theme = tt),
             ncol = 1, heights = c(0.18, 0.82))
```

Monthly Sales Data by Product (Q1)

Product_ID	Product	January	February	March	Total
1	Product A	2000	2200	2400	6600
2	Product B	1500	1800	1600	4900
3	Product C	1200	1400	1100	3700
Total		4700	5400	5100	15200

Output 3: Monthly Sales Data by Product (Table)

Observation: Product A has the highest Q1 total (\$6,600), followed by Product B (\$4,900) and Product C (\$3,700). Total Q1 sales are \$15,200, with February the best month (\$5,400).

PROGRAM 4: Interactive Product Sales Dashboard

Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

```
library(shiny)

# Dataset
sales <- data.frame(
  ProductID = 1:3,
  Product   = c("Product A", "Product B", "Product C"),
  January   = c(2000, 1500, 1200),
  February  = c(2200, 1800, 1400),
  March     = c(2400, 1600, 1100)
)
months     <- c("January", "February", "March")
prod_cols  <- c("Product A" = "#4E79A7", "Product B" = "#F28E2B",
               "Product C" = "#59A14F")

ui <- fluidPage(
  titlePanel("Product Sales Dashboard (Q1)"),
  sidebarLayout(
    sidebarPanel(width = 3,
      selectInput("products", "Select Products:",
        choices = sales$Product, selected = sales$Product,
        multiple = TRUE),
      selectInput("months", "Select Months:",
        choices = months, selected = months, multiple = TRUE)
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("groupedPlot", height = "330px")),
        column(6, plotOutput("areaPlot", height = "330px"))
      ),
      h4("Monthly Sales Table"),
      tableOutput("salesTable")
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({
    sales[sales$Product %in% input$products,
          c("ProductID", "Product", months[months %in% input$months])]
  })

  output$groupedPlot <- renderPlot({
    d <- filtered()
    req(nrow(d) > 0, ncol(d) > 2)
    m <- as.matrix(d[, -(1:2), drop = FALSE]); rownames(m) <- d$Product
    barplot(m, beside = TRUE, col = prod_cols[d$Product], ylim = c(0, 3000),
      xlab = "Month", ylab = "Sales ($)", main = "Grouped Bar Chart")
    legend("topleft", legend = d$Product, fill = prod_cols[d$Product],
      bty = "n", horiz = TRUE, cex = 0.8)
  }, width = 420, height = 320)

  output$areaPlot <- renderPlot({
    d <- filtered()
    req(nrow(d) > 0, ncol(d) > 3)
    m <- as.matrix(d[, -(1:2), drop = FALSE]); x <- seq_len(ncol(m))
    cum <- apply(m, 2, cumsum); if (is.null(dim(cum))) cum <- matrix(cum, nrow = 1)
    plot(x, cum[nrow(cum), ], type = "n", ylim = c(0, 7000), xaxt = "n",
```

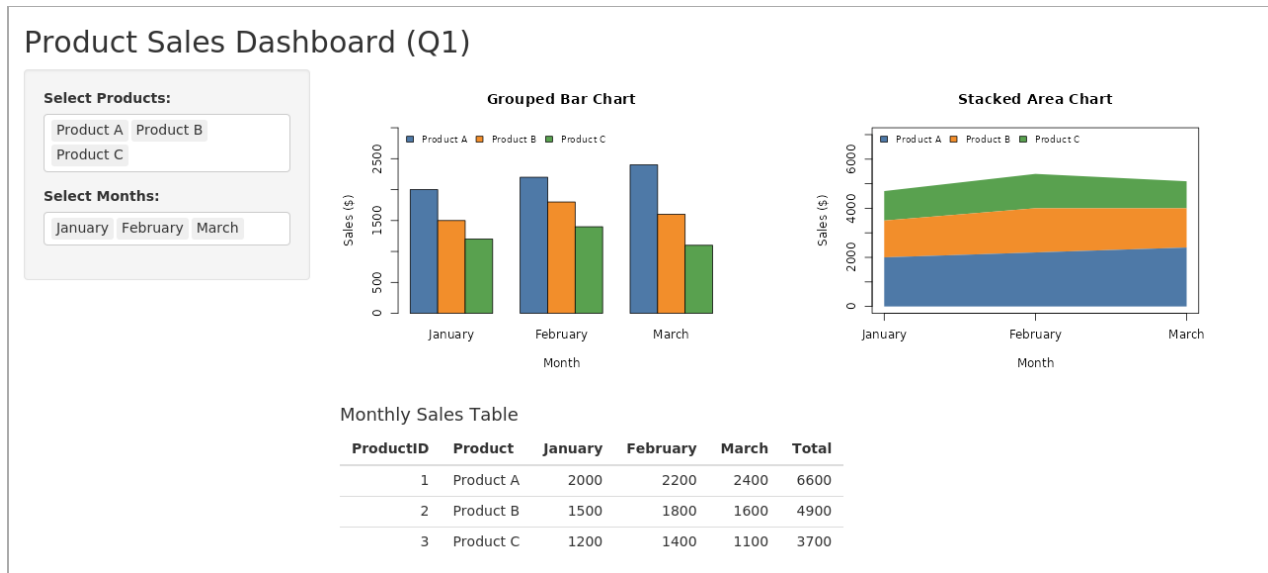
```

      xlab = "Month", ylab = "Sales ($)", main = "Stacked Area Chart")
axis(1, at = x, labels = colnames(m))
for (i in nrow(cum):1) {
  lower <- if (i == 1) rep(0, length(x)) else cum[i - 1, ]
  polygon(c(x, rev(x)), c(cum[i, ], rev(lower)),
         col = prod_cols[d$Product[i]], border = NA)
}
legend("topleft", legend = d$Product, fill = prod_cols[d$Product],
      bty = "n", horiz = TRUE, cex = 0.8)
}, width = 420, height = 320)

output$salesTable <- renderTable({
  d <- filtered()
  req(nrow(d) > 0)
  d$Total <- rowSums(d[, -(1:2), drop = FALSE])
  d
}, digits = 0)
}

shinyApp(ui = ui, server = server)

```



Output 4: Interactive Product Sales Dashboard (R Shiny)

Observation: The dashboard combines the grouped bar chart, stacked area chart and sales table on one screen. Selecting products or months updates all three views together, so the user can, for example, compare only Products A and B or focus on February.

Note: The question specifies Tableau; the equivalent dashboard was built with R Shiny. In Tableau: reshape the data to Product, Month and Sales columns, create a grouped bar sheet, an area chart sheet (Month vs SUM(Sales), Product on Color) and a text-table sheet, place all three on a dashboard, and add Product and Month filters applied to all sheets.

Result

The monthly product sales data was successfully visualized using a grouped bar chart, stacked area chart and table, and an interactive dashboard combining all three was created.